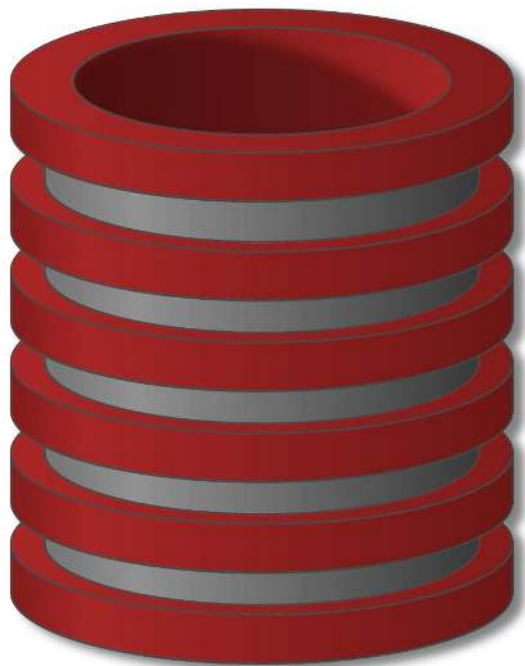




XML Data

Introduction,
Well-formed XML

Extensible Markup Language (XML)

- Standard for data representation and exchange
- Document format similar to HTML
 - Tags describe content instead of formatting
- Also streaming format

```
<?xml version="1.0" ?>
<!-- Bookstore with no DTD -->
- <Bookstore>
- <Book ISBN="ISBN-0-13-713526-2" Price="85" Edition="3rd">
  <Title>A First Course in Database Systems</Title>
  - <Authors>
    - <Author>
      <First_Name>Jeffrey</First_Name>
      <Last_Name>Ullman</Last_Name>
    </Author>
    - <Author>
      <First_Name>Jennifer</First_Name>
      <Last_Name>Widom</Last_Name>
    </Author>
  </Authors>
</Book>
- <Book ISBN="ISBN-0-13-815504-6" Price="100">
  <Remark>Buy this book bundled with "A First Course" -- a great deal!</Remark>
  <Title>Database Systems: The Complete Book</Title>
  - <Authors>
```

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    <Remark>Buy this book bundled with "A First Course" -- a great deal!</Remark>
    <Title>Database Systems: The Complete Book</Title>
    <Authors>
      <Author>
        <First_Name>Hector</First_Name>
        <Last_Name>Garcia-Molina</Last_Name>
      </Author>
      <Author>
        <First_Name>Jeffrey</First_Name>
        <Last_Name>Ullman</Last_Name>
      </Author>
      <Author>
        <First_Name>Jennifer</First_Name>

```

Handwritten annotations on the XML code:

- Arrows point to the opening tags of `<Bookstore>`, `<Book>`, `<Author>`, and `<Remark>`.
- The attribute `Edition="3rd"` is circled in pink.
- The `<Remark>` element is circled in pink.
- The `<Author>` elements for the first book are circled in pink.
- The `<Author>` elements for the second book are circled in pink.
- Handwritten text "No remark" with an arrow pointing to the `<Remark>` element.
- Handwritten text "No edition" with an arrow pointing to the `Edition="3rd"` attribute.
- Handwritten numbers 2 and 3 are circled in pink.

Basic constructs

- Tagged elements (nested)
- Attributes
- Text

Relational Model versus XML

Well-Formed XML

	Relational	XML
Structure	Tables	Hierarchical <u>Tree</u> , graph
Schema	Fixed in advance	Flexible "Self-describing"
Queries	Simple, nice langs. 😊	Less so. 😐
Ordering	None.	Implied.
Implementation	Native.	Add-on.

“Well-Formed” XML

Adheres to basic structural requirements

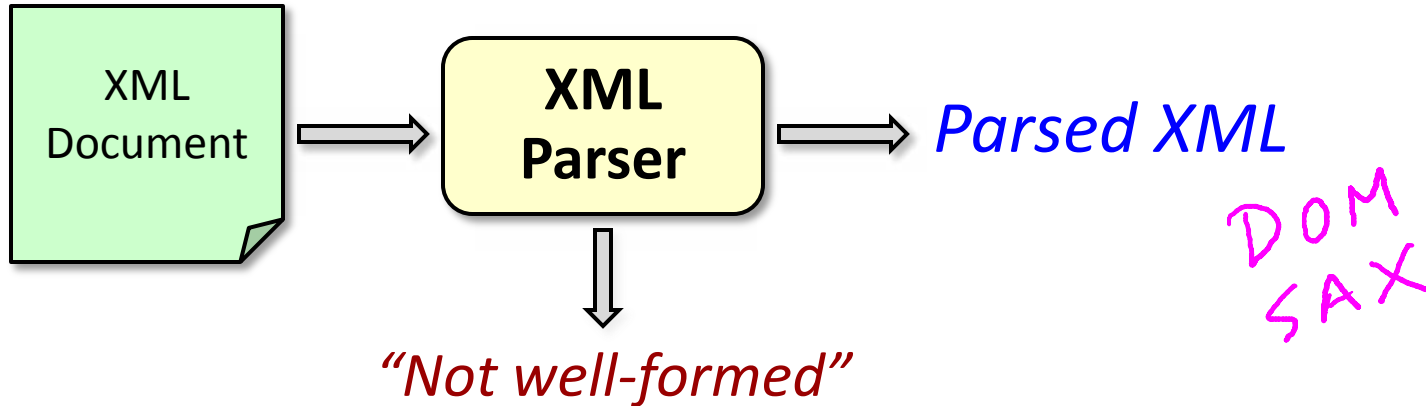
- Single root element
- Matched tags, proper nesting
- Unique attributes within elements

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“Well-Formed” XML

Adheres to basic structural requirements

- Single root element
- Matched tags, proper nesting
- Unique attributes within elements



Displaying XML

Use rule-based language to translate to HTML

- *Cascading stylesheets (CSS)*
- *Extensible stylesheet language (XSL)*

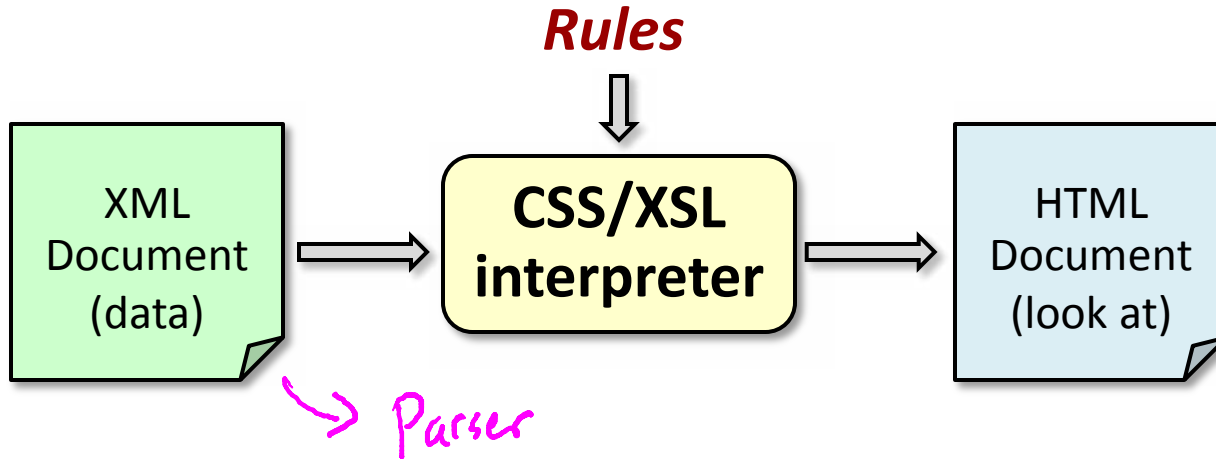


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Displaying XML

Use rule-based language to translate to HTML

- *Cascading stylesheets (CSS)*
- *Extensible stylesheet language (XSL)*



Extensible Markup Language (XML)

- Standard for data representation and exchange
- Formal specification is enormous; we cover most important components

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